

Adarsh Kumar Thakur

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SKILLS

Skills: SystemVerilog, Verilog, VHDL, UVM, Advanced Digital Design, C, C++, Communication Protocol, Assertion & Code Coverage, Static Timing Analysis, Computer Architecture

Tools: Vivado, QuestaSim, ActiveHDL, LTspice, ModelSim

WORK EXPERIENCE

CDAC Chennai || R&D

Apr 2024 - Present

Project Engineer

Design and Verification of FIDO2 Authenticator

- Designed and implemented a hardware crypto engine integrating SHA-256, HMAC, AES (GCM/ECB/CBC/CTR), ECDSA, and TRNG modules in Verilog.
- Integrated the crypto core with FPGA and SoC platforms for secure authentication flow.
- Performed FPGA synthesis, STA, and on-board validation.
- Packaged RTL as Vivado IP and integrated AXI, UART, GPIO, timers, and memory.
- Integrated the core with a RISC-V SoC and executed HW-SW co-verification.
- Verified AXI/AXI4-Stream using constrained-random testing and achieved coverage closure.
- Built SV environment with SVA, scoreboards, and functional coverage.
- Performed system-level validation and hardware testing for FIDO-based secure operations.

CDAC Hyderabad

Sep 2023 - Feb 2024

VLSI Trainee

- Specialized in Digital Design, Computer Architecture, SoC design, C, and Python with hands-on experience in Verilog, SystemVerilog, and FPGA tools.
- Performed end-to-end AXI4-Stream verification by developing a SystemVerilog-based testbench.

EDUCATION

Chandigarh University, University Institute of Engineering

June 2023

B.E. in Electronics and Communication Engineering

CGPA: 7.11/10

Relevant Coursework: VLSI Design, Digital System Design, Computer Architecture, Low Power VLSI,

Verilog HDL, Signal and System, Microprocessors and Microcontrollers

Chhotu Ram Rural Institute of Technology, Delhi

June 2020

Diploma in Electronics and Communication Engineering

CGPA: 7.00/10

PROJECT WORK

- **UVM-Based AXI4-Stream VIP (Jan 2024 - Feb 2024):** Developed a configurable AXI4-Stream VIP supporting both master and slave modes. Built a UVM-based SystemVerilog testbench with assertions and functional coverage. Verified protocol compliance using constrained-random testing and coverage analysis.
- **Gas Sensor System (Feb 2023 - June 2023):** Designed a gas detection system with high sensitivity and selectivity. Implemented signal processing and alarm indication features. Achieved reliable detection performance with optimized response time.
- **Head-Tilt Controlled Wheelchair (Aug 2022 - Dec 2022):** Developed a control system using head-motion sensing for wheelchair navigation. Translated head tilt angles into directional movement commands. Integrated safety features including speed control and emergency stop.

COURSES AND CERTIFICATES

- PG - Diploma in VLSI from CDAC - Hyderabad (R&D) under MeitY
- NPTEL (Sensors and Actuators)
- TIMTS and AICRA (Robotics)
- Digital Systems: from Logic Gates to Processors
- Introduction to Electronics